Mohanna Hoveyda

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RESEARCH INTERESTS

Conversational AI, Reinforcement learning for multi-agent information systems, Information retrieval, LLM-based reasoning.

EDUCATION

PhD, Radboud University, iCIS Nijmegen, The Netherlands
Knowledge-Enriched Conversational AI *Feb 2023–Feb 2027 (anticipated)*
Advisors: Arjen P. de Vries, Faegheh Hasibi, Maarten de Rijke
Research Group: Informagus

MSc, University of Tehran, FNST Tehran, Iran
Computational Linguistics *Sep 2020–Jan 2023*
Thesis: Network Science-Based Structural Analysis of Human- and Machine-Based Semantic Representations

Bachelor, University of Tehran, ECE & FFLL Tehran, Iran
Computer Engineering (Minor) *2018–2020*
French Language and Literature (Major) *2016–2020*

RESEARCH

OrLog: Resolving Complex Queries with LLMs and Probabilistic Logic Programming **Under Review**
Mohanna Hoveyda, Jelle Piepenbrock, Arjen P. de Vries, Maarten de Rijke, Faegheh Hasibi

- Built a neuro-symbolic retrieval framework that integrates *probabilistic priors from LLMs* with a *logic programming engine* to enforce query constraints (i.e., conjunction, disjunction, negation) in ranking.
- Achieved higher accuracy with ∼90% lower token cost compared to end-to-end LLM reasoning, demonstrating how smaller models can replace brute-force scaling for large-scale retrieval systems.

Adaptive Orchestration of Modular Generative Information Access Systems **SIGIR 2025**
Mohanna Hoveyda, Harrie Oosterhuis, Arjen P. de Vries, Maarten de Rijke, Faegheh Hasibi

- Designed a modular GenIA architecture for real-time adaptive orchestration of *LLM agents*, retrieval models, and external tools per query to maximize performance while cutting computational overhead.
- Developed a dynamic QA system that adaptively selects optimal answering strategies based on question complexity under resource constraints, using *contextual multi-armed bandits*.

Real-World Conversational Entity Linking Requires More Than Zero-Shots **ACL 2024 Findings**
Mohanna Hoveyda, Arjen P. de Vries, Maarten de Rijke, Faegheh Hasibi

- Built a Reddit-based conversational EL dataset to evaluate zero-shot models on unseen, domain-specific knowledge bases.
- Fine-tuned and evaluated BERT-based EL models, revealing severe accuracy drops on unseen knowledge bases.

Network Science-Based Structural Analysis of Heterogeneous Semantic Representations **Master's Thesis**

Mohanna Hoveyda, Paulino Villas-Boas, Mahmood Bijankhan, Mostafa Salehi

- Designed a statistical framework with configuration models and hypothesis testing for size-invariant comparison and clustering of non-isomorphic networks.
- Applied graph-theoretic metrics to compare semantic representations across machine-learned (uni-modal: Word2Vec, BERT; multi-modal: VisWord2Vec), human-curated (WordNet, ConceptNet), and hybrid (ConceptNet Numberbatch) spaces.

PRESENTATIONS

Adaptive Orchestration of Agents, Tool Usage, and Reasoning in Information Access Systems. Oral presentation, DIR'25.

Designs of Future Information Access Systems Should Be Modular, Adaptive and Dynamic. Poster presentation, CONSEQUENCES at RecSys'25

Understanding Diverse Reasoning Procedures in Foundation Models via Mechanistic Interpretability. Poster presentation, Cognitive Computational Neuroscience Conference (CCN'25).
Extended abstract co-authored with Jasmin Kareem, Roxana Petcu, Angela van Sprang, and Ana Lucic.

Adaptive Orchestration of Modular Generative Information Access Systems. Oral presentation, SIGIR'25.

Real-World Conversational Entity Linking Requires More Than Zero-Shots. Poster presentation, ACL'24.

SKILLS & TOOLS

Python, C, C++
PyTorch, TensorFlow, Scikit-learn, Hugging Face
ElasticSearch, Lucene, Faiss
HPC, vLLM
Z3 SMT solver, ProbLog
NetworkX, Matplotlib, Seaborn

LANGUAGES

Persian (Native), *English* (Fluent), *French* (Intermediate), *Dutch* (Basic)

TEACHING & SUPERVISION

Teaching Assistant Information Retrieval (Radboud University), Machine Learning for NLP (University of Tehran)

Supervision

Tsvetomira Krikoryan (MSc internship, Completed, Wintertuin Company):
Training a GPT-2 model on Papiamentu for creative task assistance

Luuk van den Hurk (MSc thesis, September 2025, KPN Company):
Enhancing Knowledge Representation for more effective RAG

Mario Tsatsev (MSc thesis, September 2025, Radboud University):
Enhanced prior probability estimation for neuro-symbolic reasoning

AWARDS & SCHOLARSHIPS

Ranked 6th in Computational Linguistics National Master's Entrance Exam, Iran National Organization for Educational Testing (2020)

Scholarship for Computational Linguistics Master's Program, University of Tehran, Iran (2020)

WORKSHOPS & SCHOOLS

European Summer School on Information Retrieval (ESSIR), University of Amsterdam, Netherlands (2024); Technische Universität Wien, Austria (2023)

High-Performance Computing (HPC) and Deep Learning Workshop, HPC Laboratory, IPM, Tehran, Iran (2022)

Summer School of Intelligent Learning, IPM & Sharif University of Technology, Tehran, Iran (2019)

Autumn School of Mind and Brain, Cognitive Sciences and Technologies Council & University of Tehran, Tehran, Iran (2018)

REFERENCES

Arjen P. de Vries `a.devries@cs.ru.nl`
Professor of Information Retrieval at Radboud University and Research Director of iCIS

Maarten de Rijke `m.derijke@uva.nl`
Professor of Artificial Intelligence and Information Retrieval at UvA